Question 1:

A data center implements a multi-layered physical access control system. The probability of an unauthorized individual breaching each layer is as follows:

- **Perimeter fence:** 5%
- **Security guards:** 3%
- **Biometric access control:** 1%
- **Mantrap entry system:** 0.5%

Assuming each layer operates independently, calculate the overall probability that an unauthorized individual can access the data center.

Solution:

The overall probability (P_total) of unauthorized access is the product of the probabilities of breaching each layer:

P_total = P_fence × P_guards × P_biometric × P_mantrap

Converting percentages to decimals:

- P_fence = 0.05
- P_guards = 0.03
- P_biometric = 0.01
- P_mantrap = 0.005

Calculating the product:

P_total = 0.05 × 0.03 × 0.01 × 0.005 = 7.5 × 10^-9

Converting back to a percentage:

P_total = 7.5 × 10^-7%

Therefore, the overall probability of unauthorized access is 0.00000075%.

Question 2:

An organization has determined that the annual rate of occurrence (ARO) for unauthorized physical access attempts is 4. The estimated loss per incident (SLE) is $150,000. Calculate the Annual Loss Expectancy (ALE) for unauthorized physical access incidents.

Solution:

Annual Loss Expectancy (ALE) is calculated using the formula:

ALE = ARO × SLE

Given:

- ARO = 4
- SLE = $150,000

Calculating ALE:

ALE = 4 × $150,000 = $600,000

Therefore, the Annual Loss Expectancy is $600,000.

Question 3:

A company is evaluating the cost-effectiveness of installing a new biometric access control system. The system costs $80,000 to install and reduces the probability of unauthorized access from 2% to 0.5%. If the potential loss from a single unauthorized access incident is $500,000 and the annual rate of occurrence is 3, determine whether the investment is justified over a one-year period.

Solution:

First, calculate the ALE before and after installing the biometric system.

Before installation:

- Probability of unauthorized access (P_before) = 2% = 0.02
- Annual rate of occurrence (ARO) = 3
- Single Loss Expectancy (SLE) = $500,000

ALE_before = ARO × P_before × SLE

ALE_before = 3 × 0.02 × $500,000 = $30,000

After installation:

- Probability of unauthorized access (P_after) = 0.5% = 0.005

ALE_after = ARO × P_after × SLE

ALE_after = 3 × 0.005 × $500,000 = $7,500

Cost-Benefit Analysis:

- Reduction in ALE = ALE_before - ALE_after = $30,000 - $7,500 = $22,500

- Cost of biometric system = $80,000

Since the cost of the system ($80,000) exceeds the annual savings ($22,500), the investment is not justified over a one-year period.

Question 4:

In a facility, the access logs show an average of 250 authorized entries per day. The security policy mandates that no more than 0.1% of entries should be unauthorized. Calculate the maximum number of unauthorized entries allowed per day according to the policy.

Solution:

Maximum allowed unauthorized entries = Total entries × Unauthorized entry rate

= 250 × 0.001 = 0.25

Since the number of entries must be a whole number, the facility should aim for 0 unauthorized entries per day to comply with the policy.

Question 5:

A security audit reveals that the response time to a physical security breach is critical in minimizing losses. The current average response time is 15 minutes, resulting in an average loss of $10,000 per incident. If the response time is improved to 5 minutes, the average loss per incident is expected to decrease proportionally. Calculate the expected average loss per incident with the improved response time.

Solution:

Assuming the loss is directly proportional to the response time:

Current response time = 15 minutes Current average loss = $10,000

Improved response time = 5 minutes

Loss reduction factor = Improved response time / Current response time

= 5 minutes / 15 minutes = 1/3

Expected average loss with improved response time = Current average loss × Loss reduction factor